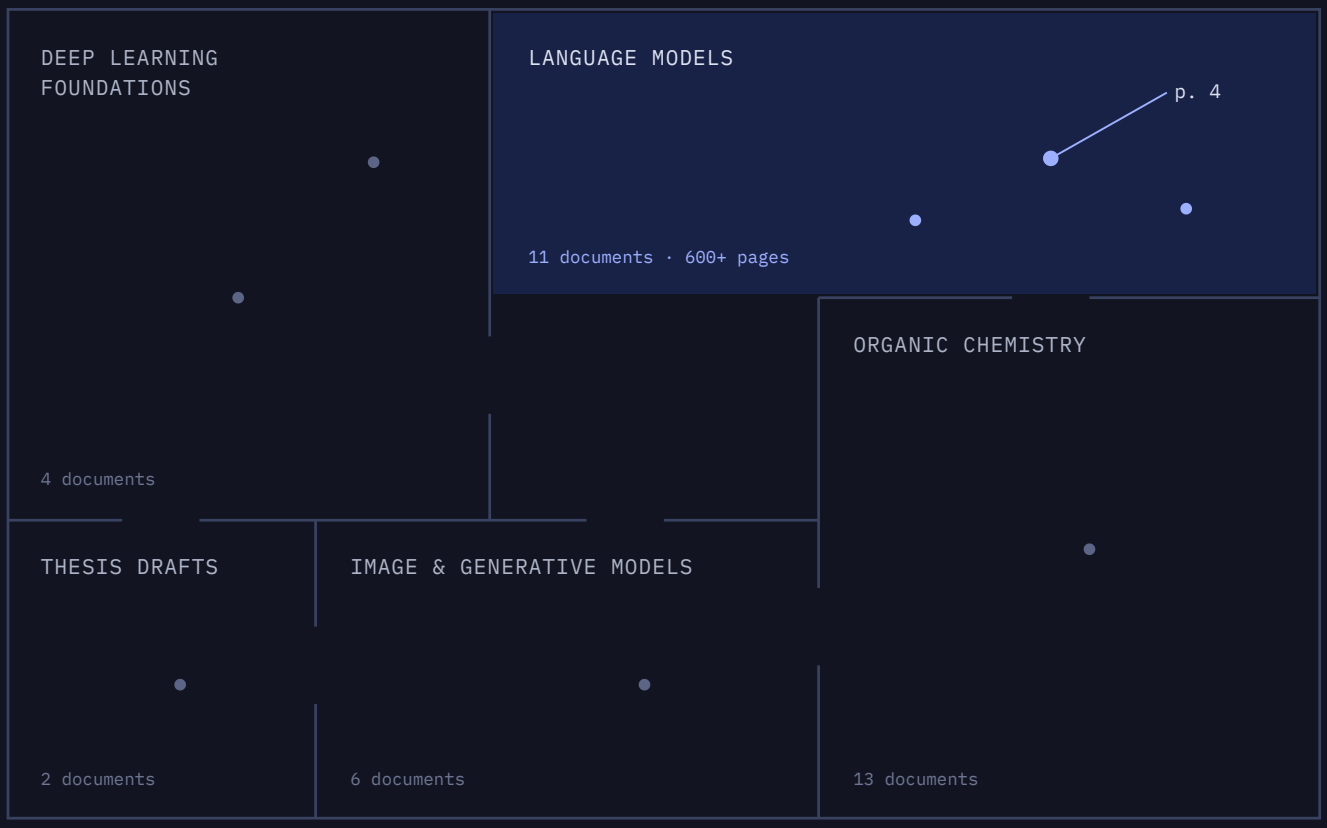
A PRIVATE KNOWLEDGE BANK

Loci

Your knowledge, kept in its place.

A desktop app that turns everything you read (papers, books, lecture notes) into a library you can search by words or by meaning, with every result pointing to its page. It runs entirely on your own computer.



WHY WE ARE BUILDING IT

You read hundreds of pages. Finding the one you need *should take seconds.*

Students and researchers live in documents: papers, textbooks, lecture notes, past exams. The tools around them were not built for collections that keep growing.

"I know I read it somewhere."

Search inside one PDF at a time, or re-read whole chapters to find a passage you saw last week.

"I uploaded the same papers again."

Online AI tools forget everything between sessions, and every upload means handing your documents to someone else's server.

"Where did that answer come from?"

AI answers sound confident, but checking them against the source means hunting for it yourself.

The idea: a knowledge bank that stays with you

Import once — Add files or whole folders. Loci reads them in the background and keeps them, organized into workspaces.

Find anything — Search by exact words or by meaning, across everything, and jump straight to the page.

Ask — Get answers written only from your documents, with every claim cited to a page you can open. NEXT

Keep it private — Documents and AI stay on your computer. No account, no uploads, works offline.

HOW IT WORKS

One place per subject. *Everything inside it, findable.*

The name comes from the method of loci, the memory palace: you remember things by the place you keep them. In Loci each subject, project or book gets its own workspace, and every result leads back to its page.

STEP 1

Create a workspace

One per course, research project or book, so collections never mix. The same file can live in several.

AVAILABLE

STEP 2

Add documents

Drop in files or whole folders: PDF, text, Markdown. Duplicates are spotted, and small files come first.

AVAILABLE

STEP 3

Loci reads them

Text is extracted page by page and indexed in the background. Word search works within seconds.

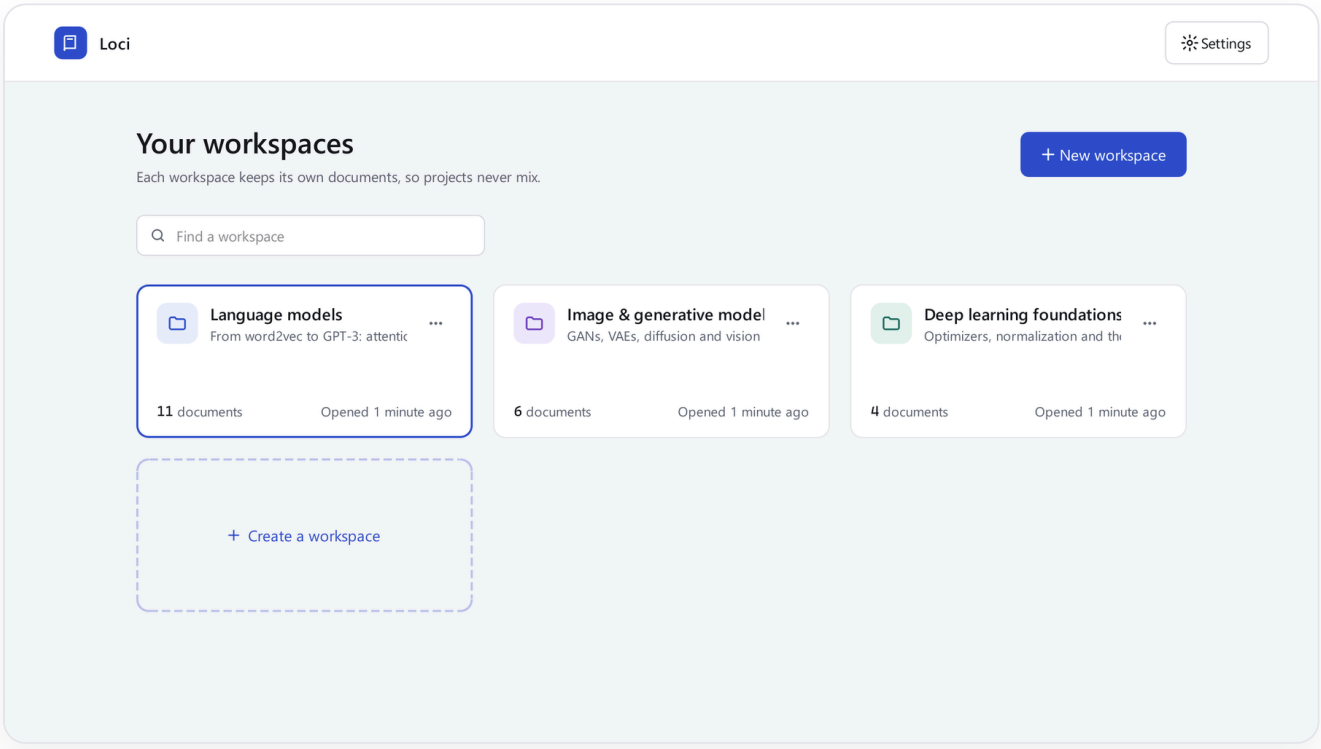
AVAILABLE

STEP 4

Find, then ask

Search by words or meaning today; ask questions and get cited answers next.

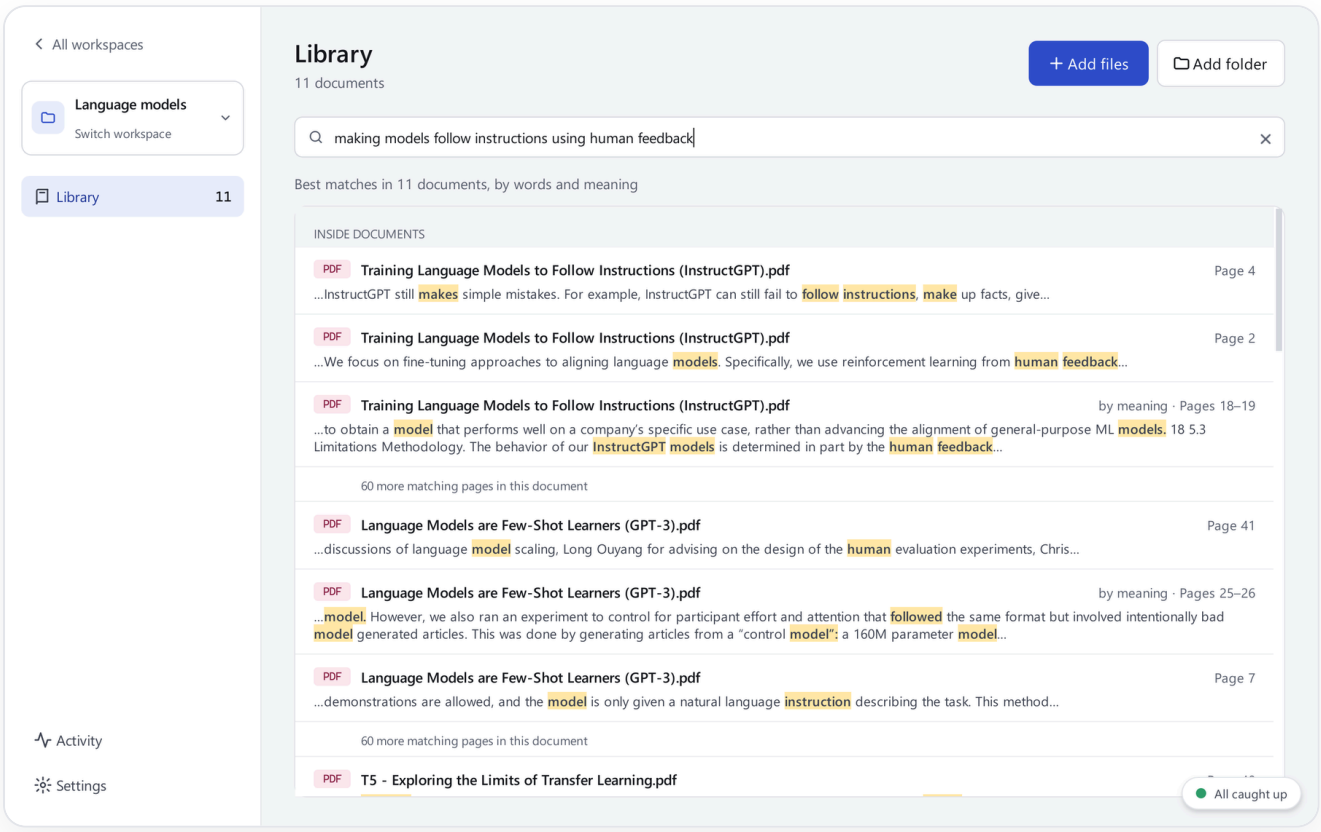
ASK: NEXT



The home screen: one workspace per subject. Real screenshot, version 0.3, with 21 machine-learning papers.

FIND ANYTHING AVAILABLE NOW

Search the way you think, *not the way it was written.*



"making models follow instructions using human feedback": the InstructGPT paper first, with matching passages and their pages. Real screenshot, version 0.3.

By words

Exact terms, names and formulas, with word forms matched (*learn* finds *learning*). Matches are highlighted.

By meaning

Describe an idea in your own words. A small model running on your computer finds passages that say it differently.

Both, combined

The two rankings are merged: the precision of words and the reach of meaning in one list.

Straight to the page

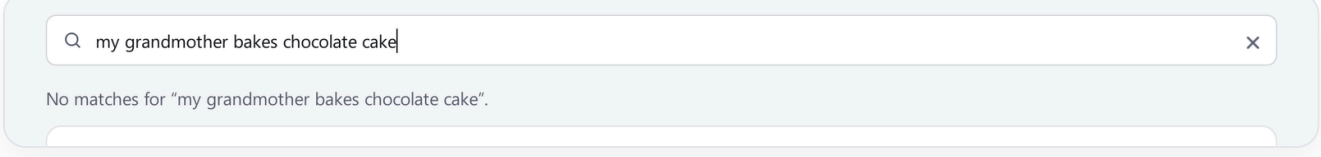
Every result names its page. One click opens the document there.

Only real matches

A page must actually support your search to be shown. If nothing does, Loci says so.

Fast at any size

Searches answer in under a tenth of a second, even across hundreds of documents.

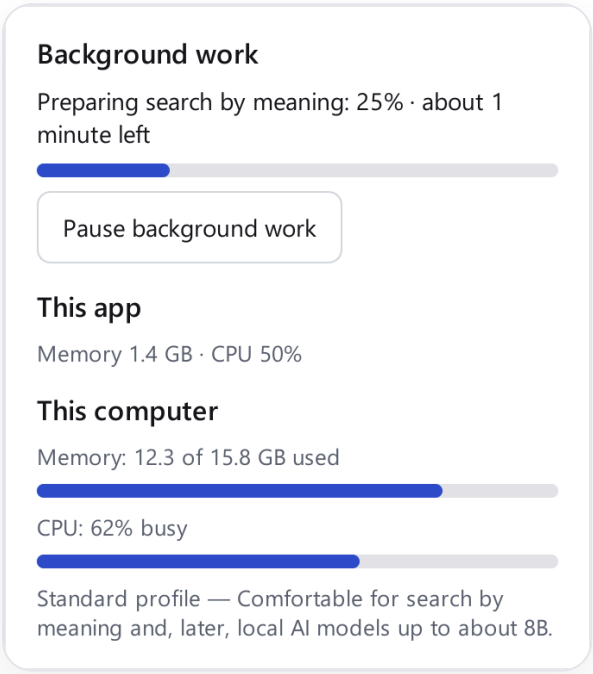


Unrelated text finds nothing. Loci never pads results with guesses.

BUILT FOR ORDINARY LAPTOPS AVAILABLE NOW

No graphics card, no cloud. *Just your computer.*

Loci is designed for the laptops students and researchers already own (8 to 16 GB of memory) and stays responsive while it works.



The status panel, always one click away.

- **Searchable first, smarter later**

  Documents are word-searchable within seconds of import. Search by meaning is prepared in the background, with the time left shown.

- **Knows the computer it runs on**

  Loci detects how much memory you have and picks lighter or stronger local models to match.

- **Polite in the background**

  Pause and resume at any time. Loci pauses on its own when memory runs low, and continues where it stopped after a restart.

- **Offline after setup**

  The only download is a 90 MB search model, once. After that, nothing leaves your computer.

87%	0.14 s	<0.1 s	~30 min
of test questions found the right paper in the top 5 results	to read a typical PDF (median)	for a search across all documents	to prepare search by meaning for 358 papers, in the background

Measured on 358 openly licensed arXiv papers (9,972 pages) with 45 test questions, on a 16 GB laptop without a graphics card.

ASK YOUR DOCUMENTS COMING IN V0.4

Answers you can check, *page by page.*

The Ask section will answer questions using only your documents. Each claim is checked against the passage it came from and cited to a page you can open. When your documents don't contain the answer, Loci will say so instead of guessing.

DESIGN PREVIEW · NOT IN THE CURRENT VERSION

Language models · Ask — Answers from 11 documents

How does LoRA make fine-tuning large models cheaper?

LoRA keeps the pre-trained weights frozen and trains only a pair of small low-rank matrices added to each layer `LoRA · p. 1`. Because the number of trainable parameters drops sharply, fine-tuning needs far less GPU memory, and each task only has to store the small matrices `LoRA · p. 2`. Unlike adapter layers, the matrices can be merged into the weights after training, so answers are not slower to produce `LoRA · p. 4`.

Every sentence checked against its source passage

LoRA: Low-Rank Adaptation
page 1 · open at page

LoRA: Low-Rank Adaptation
page 2 · open at page

LoRA: Low-Rank Adaptation
page 4 · open at page

Which GPU did the authors use?

> The documents in this workspace don't say which GPU the LoRA authors used, so Loci can't answer that from them. The closest passage describes the memory saved during training `LoRA · p. 1`.

Only from your documents

The AI writes from the passages Loci found, not from what it half-remembers.

Checked, then cited

Each claim is verified against its passage before you see it. Click a citation to open the page.

Local and private

The AI model runs on your computer, sized to its memory. Nothing is sent online.

EVERYTHING LOCI WILL DO

One shared core, *plus what you need.*

Every user gets the same core. Specialized tools come as optional extensions, the way code editors work, so a chemistry student and a historian each install only what helps them.

FEATURE	WHAT IT DOES	STATUS
	Organize and find	
Workspaces	Separate collections per subject or project	NOW
Import files and folders	PDF, text and Markdown; duplicates detected; progress and time left	NOW
Search by words and meaning	Combined ranking, highlighted passages, page numbers	NOW
Open at page	Jump from any result to its page in your PDF viewer	NOW
Background status	Time left, computer load, pause, automatic low-memory pause	NOW
	Understand	
Ask with citations	Answers from your documents, checked and cited to pages	V0.4
Conversation memory	Each workspace remembers past questions; memory you can see and edit	V0.4
Summaries	A short overview of each document, and summaries on request	V0.5
	Study and analyze	
Flashcards and quizzes	Generated from your material, with links back to the source	V0.6
Exam-paper analysis	Question patterns, topics and difficulty across past papers; practice papers in the same pattern	V0.6
Knowledge packs	Installable offline knowledge such as dictionaries and reference data	V0.7
Suggestions	Related documents, recurring topics, "compare these papers?"	V0.8
	Extensions	
More formats	Scanned PDFs (text recognition), Word, PowerPoint, EPUB	PLANNED
Subject tools	Mathematics, chemistry, engineering: formulas, tables, calculators the AI can use	PLANNED
Styled outputs	Designed flashcards, formatted summaries and reports	PLANNED

WHAT WE STAND FOR

Trustworthy by design, *built in the open.*

i Private by default

Your documents and the AI stay on your computer. No accounts, no uploads.

ii Every answer has a source

Results and answers point to the page they came from, so you can always check.

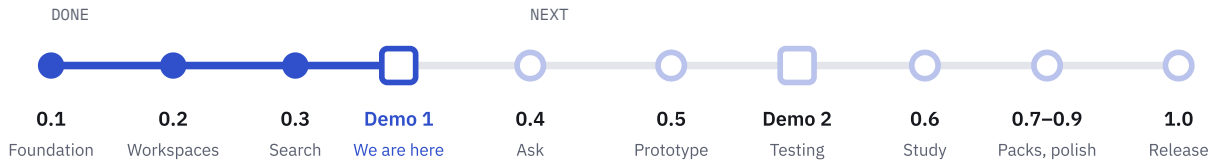
iii Made for everyday hardware

Fast on 8 to 16 GB laptops; it adapts to the computer instead of demanding a better one.

iv Tested with real users

Development pauses at planned demo gates so people outside the team can try it before we go further.

ROADMAP



Now

Version 0.3 is with testers. We are watching how people import, organize and search, and fixing what confuses them.

Next

Version 0.4 adds Ask: answers from your documents, checked and cited, from a local AI model chosen by benchmark.

After

A polished prototype, then study tools, exam-paper analysis, knowledge packs and optional extensions.

Try it

Version 0.3 is open for testing on Windows 10 and 11. Download the zip, extract it and double-click Loci.exe. No installation or Python needed.

`github.com/AzureSky0/Local-Research-Assistant-and-Study-Buddy-Hybrid/releases`

Follow the project

Loci is developed in the open as a learning project. The code, the roadmap and the decisions behind it are on GitHub. Feedback and testers are welcome.

`github.com/AzureSky0/Local-Research-Assistant-and-Study-Buddy-Hybrid`

Loci is a working name; the final name will be chosen before the public release. Screenshots show version 0.3; the Ask page is a design preview of version 0.4.